Signals and Systems Lecture-2: Continuous-Time Transformations

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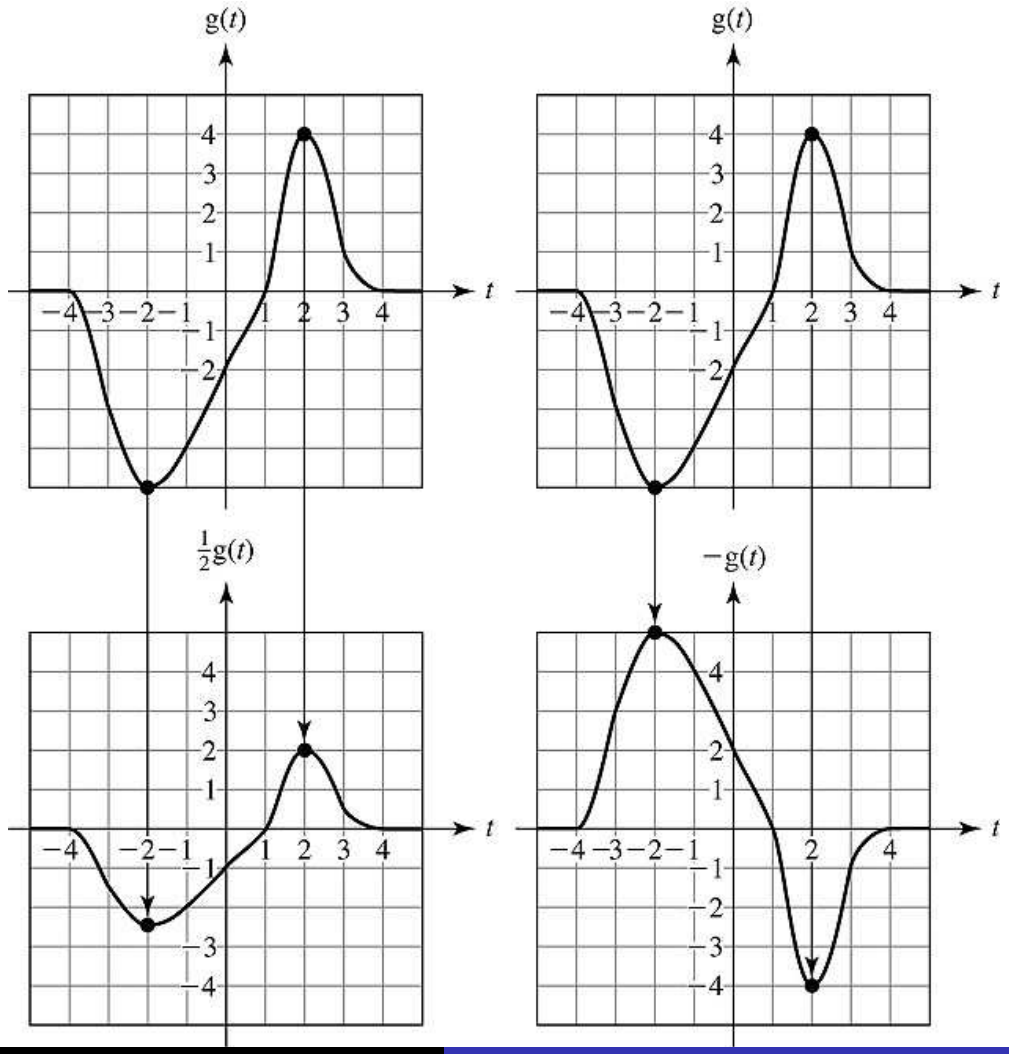
Amplitude Scaling

- Amplitude Scaling: the function $g(t)$ is multiplied by the constant A for all values of t:

$$g(t) \rightarrow Ag(t)$$

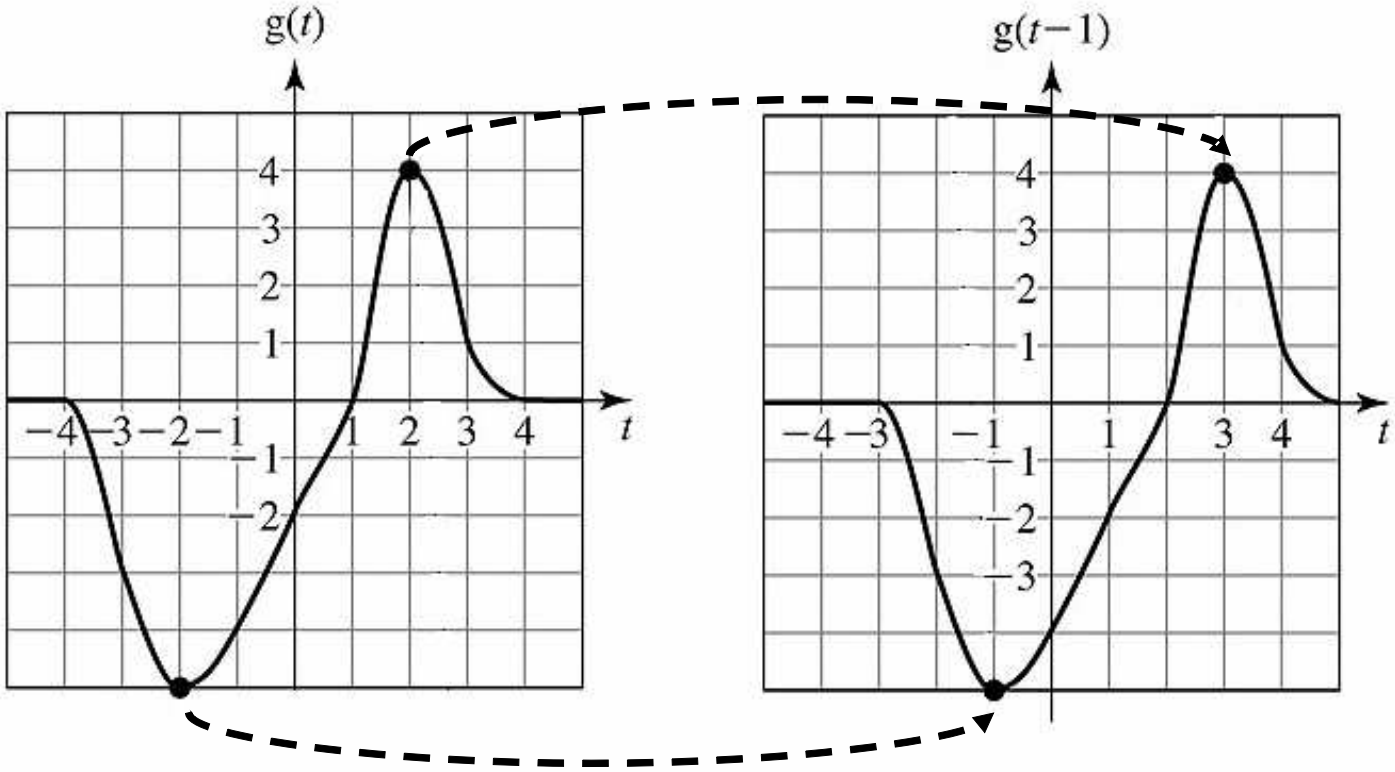


Time Shifting (1)

- Time Shifting by t_0 corresponds to the transformation $t \to t - t_0$:

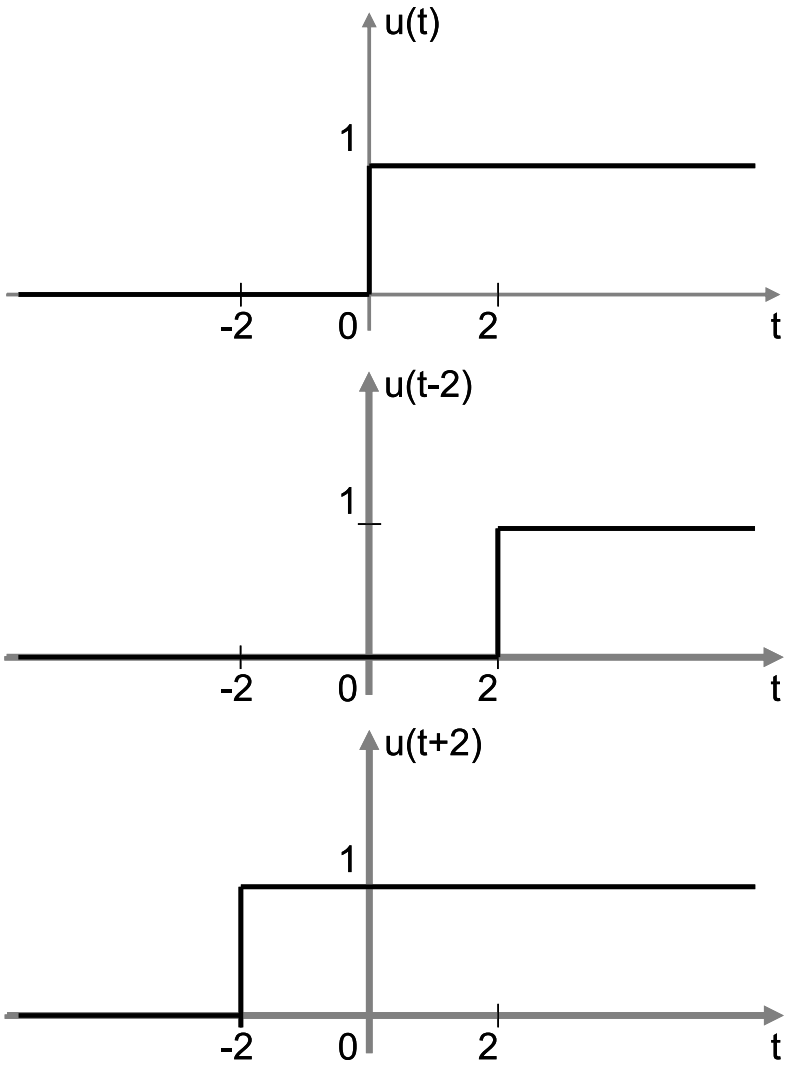
$$g(t) \to g(t - t_0)$$

- For example, the next figure shows the relation between $g(t)$ and $g(t-1)$.



Time Shifting (2)

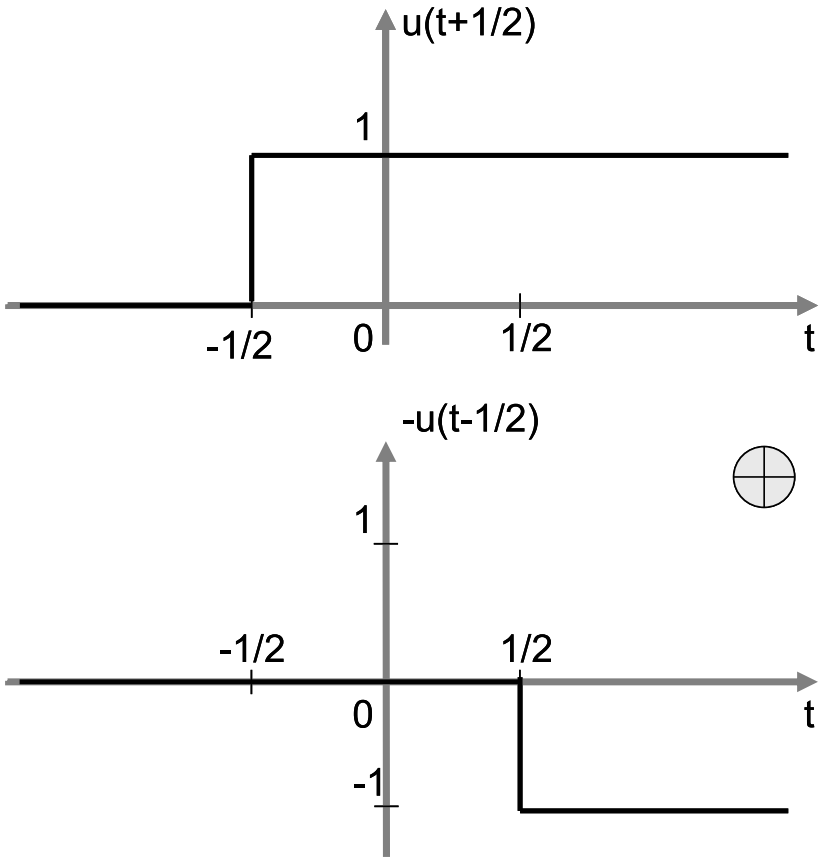
- The next figure shows $u(t-2)$ and $u(t+2)$:

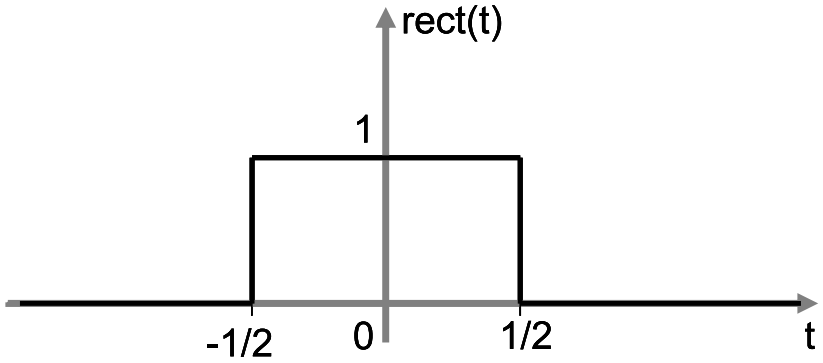


Time Shifting (3)

- The unit rectangle function can be written as:

$$\mathrm{rect}(t) = u\left(t + \frac{1}{2}\right) - u\left(t - \frac{1}{2}\right)$$

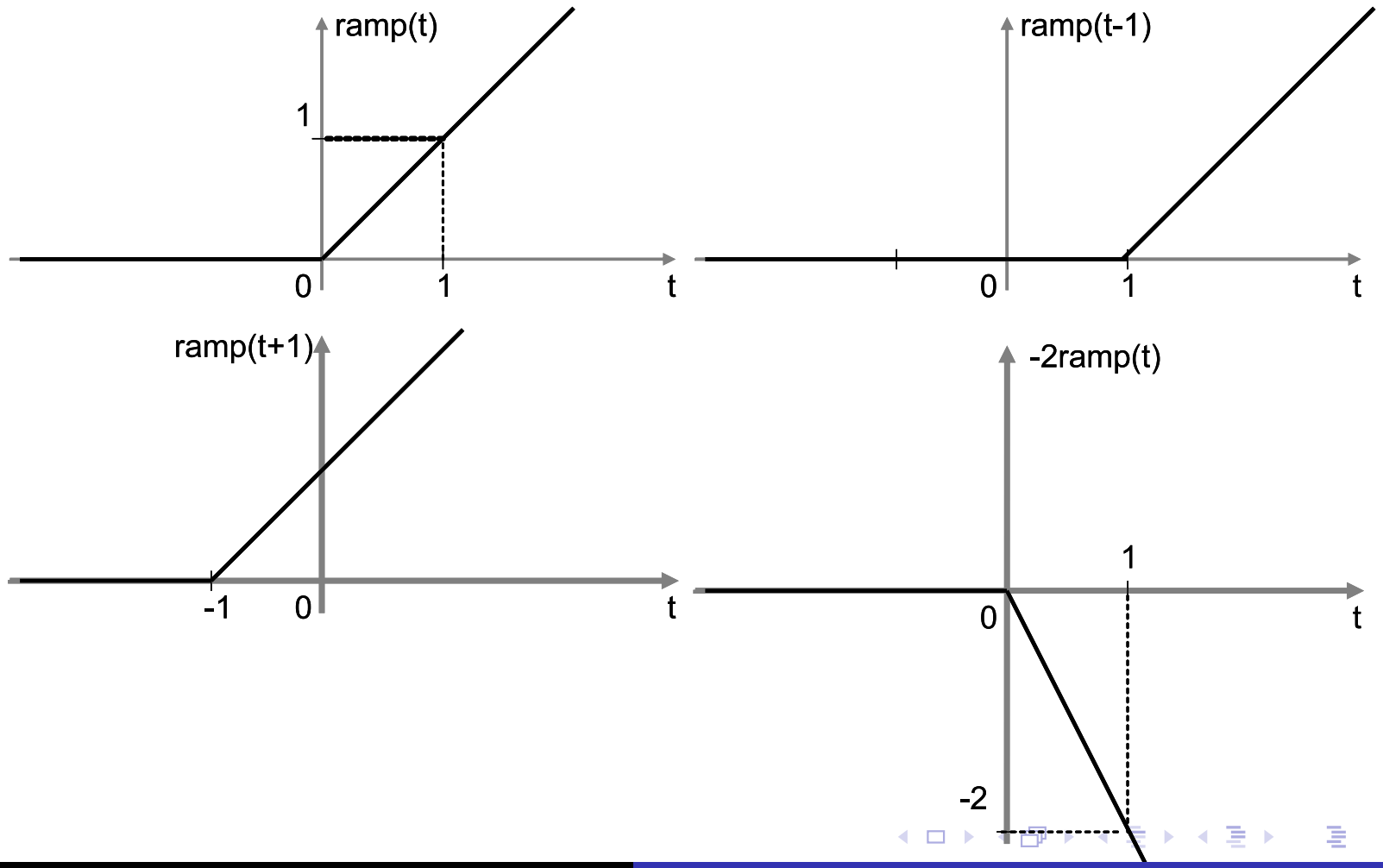




Time Shifting (4)

- The unit triangle function can be written as:

$$\mathrm{tri}(t) = \mathrm{ramp}(t+1) - 2\mathrm{ramp}(t) + \mathrm{ramp}(t-1)$$



Time Shifting (5)

- Note that:

$$\text{ramp}(t+1) = \begin{cases} t+1, & t \geq -1; \\ 0, & t < -1. \end{cases}$$

$$2\text{ramp}(t) = \begin{cases} 2t, & t \geq 0; \\ 0, & t < 0. \end{cases}$$

$$\text{ramp}(t-1) = \begin{cases} t-1, & t \geq 1; \\ 0, & t < 1. \end{cases}$$

Let $f(t) = \text{ramp}(t+1) - 2\text{ramp}(t) + \text{ramp}(t-1)$.

- For $t \leq -1$: $f(t) = 0 - 0 + 0 = 0$.
- For $-1 \leq t \leq 0$: $f(t) = t + 1 - 0 + 0 = t + 1$.
- For $0 \leq t \leq 1$: $f(t) = t + 1 - 2t + 0 = 1 - t$.
- For $t \geq 1$: $f(t) = t + 1 - 2t + t - 1 = 0$.
- Therefore: $f(t) = \text{tri}(t)$.

Time Scaling (1)

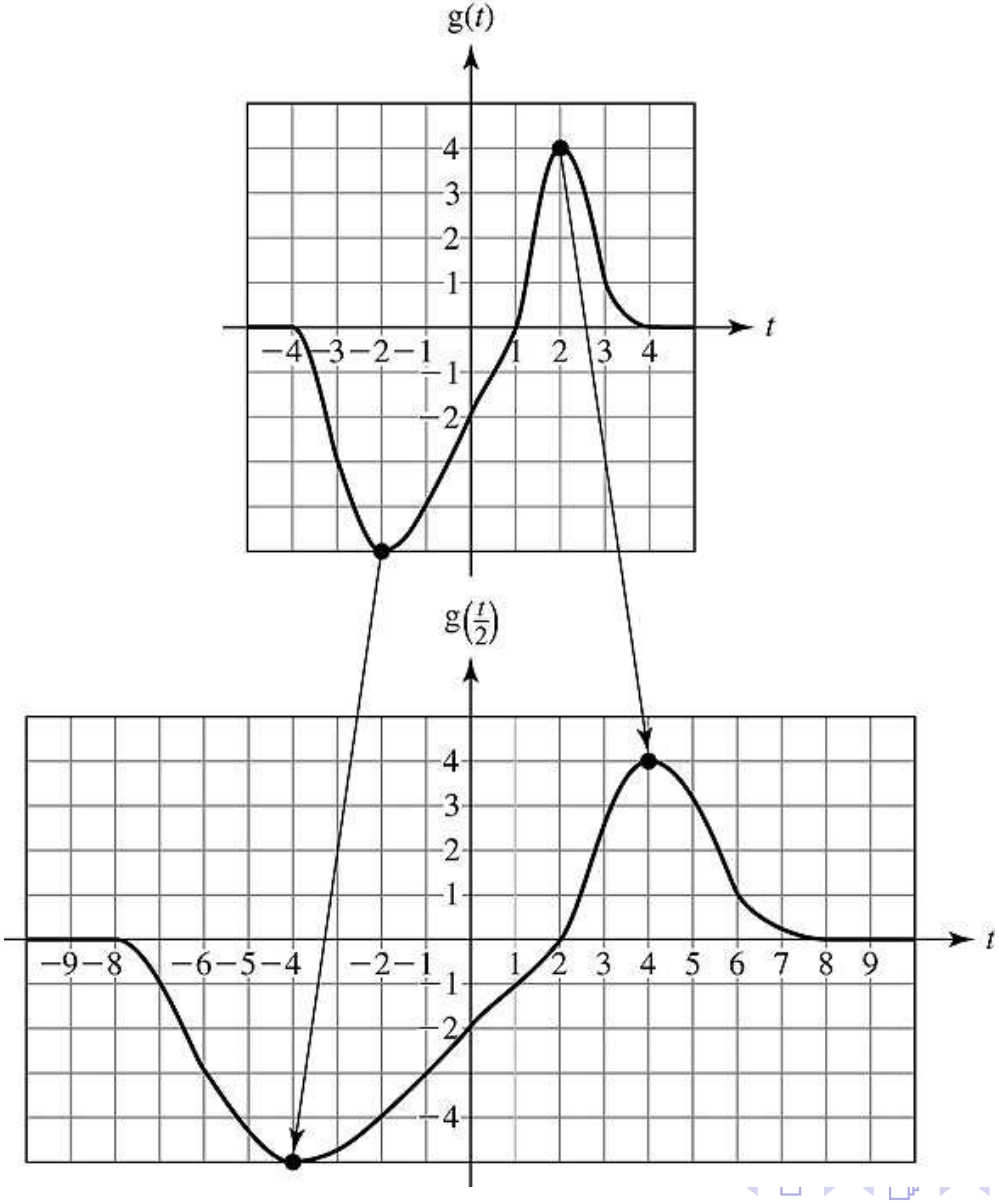
Time Scaling corresponds to the transformation $t \to t/a$:

$$g(t) \to g\left(\frac{t}{a}\right)$$

- If $a > 0$, $t \to t/a$ expands the function horizontally by a factor of a.
- If $a < 0$, $t \to t/a$ corresponds to applying the two consecutive transformations $t \to -t$ followed by $t \to t/|a|$.
  - $t \to -t$ corresponds to the time inversion that can be realized by taking the symmetry of the signal with respect to the vertical axis $t = 0$.
  - In other words, the function is first time-inverted then expanded by a factor of $|a|$.

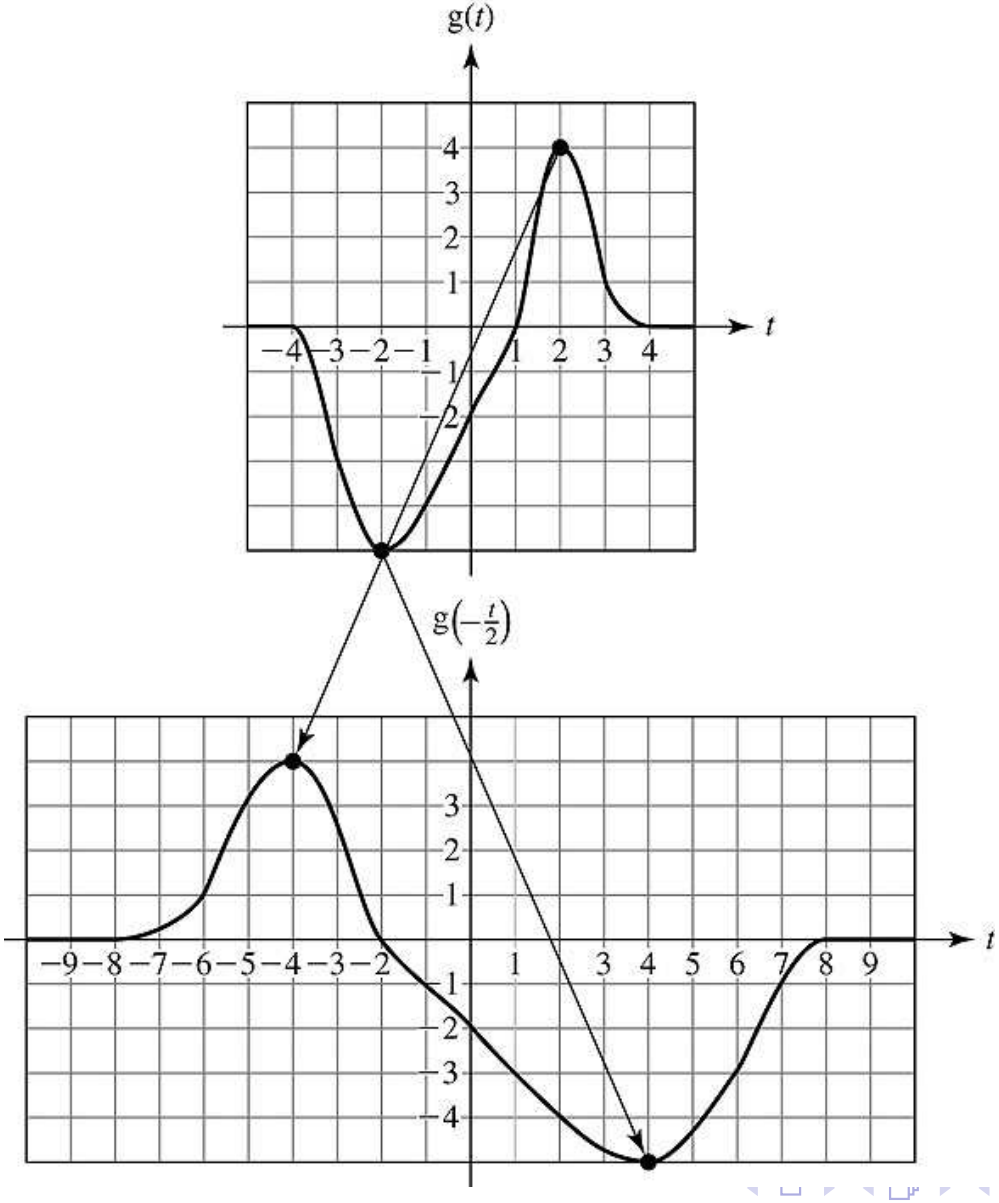
Time Scaling (2)

- The next figure compares $g(t)$ and $g\left(\frac{t}{2}\right)$:



Time Scaling (3)

- The next figure compares $g(t)$ and $g\left(\frac{-t}{2}\right)$:



Multiple Transformations (1)

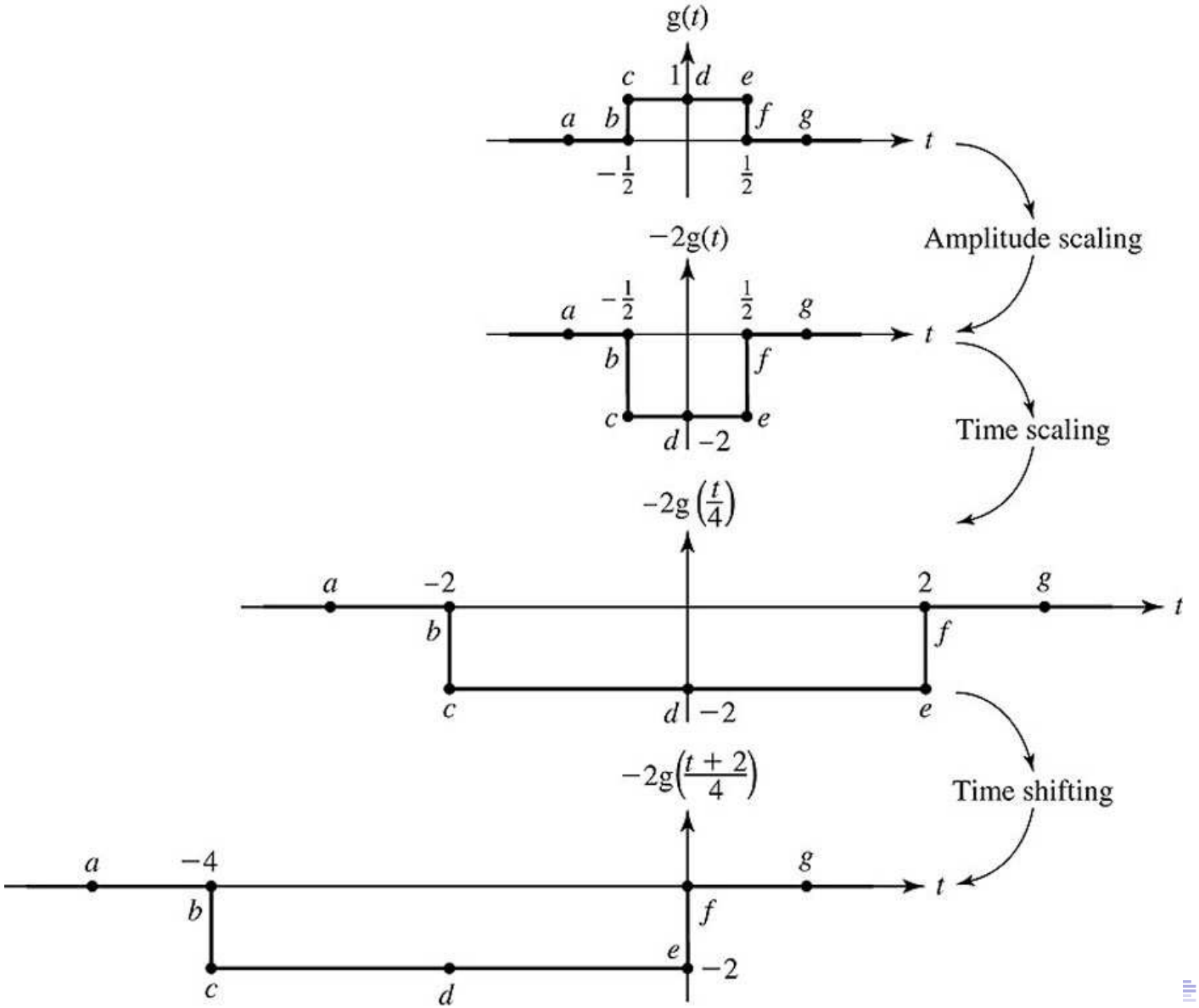
- Amplitude scaling, time shifting and time scaling can be performed simultaneously. For example:

$$g(t) \rightarrow Ag\left(\frac{t-t_0}{a}\right)$$

- Multiple transformations can be broken down into successive simple transformations:
  - Amplitude scaling: $g(t) \rightarrow Ag(t)$.
  - Time scaling $t \rightarrow t/a$: $Ag(t) \rightarrow Ag\left(\frac{t}{a}\right)$.
  - Time shifting $t \rightarrow t - t_0$: $Ag\left(\frac{t}{a}\right) \rightarrow Ag\left(\frac{t-t_0}{a}\right)$.
- Note that the order of the transformations is important. For example, a wrong answer will be obtained if we exchange the order of time-scaling and time-shifting:
  - Amplitude scaling: $g(t) \rightarrow Ag(t)$.
  - Time shifting $t \rightarrow t - t_0$: $Ag\,(t) \rightarrow Ag\,(t - t_0)$.
  - Time scaling $t \rightarrow t/a$:
    $Ag\,(t - t_0) \rightarrow Ag\left(\frac{t}{a} - t_0\right) \neq Ag\left(\frac{t-t_0}{a}\right)$.

Multiple Transformations (2)

- The next figure corresponds to $g(t) \rightarrow -2g\left(\frac{t+2}{4}\right)$:



Multiple Transformations (3)

- Consider the transformation given by:

$$g(t) \rightarrow Ag\left(bt - t_0\right)$$

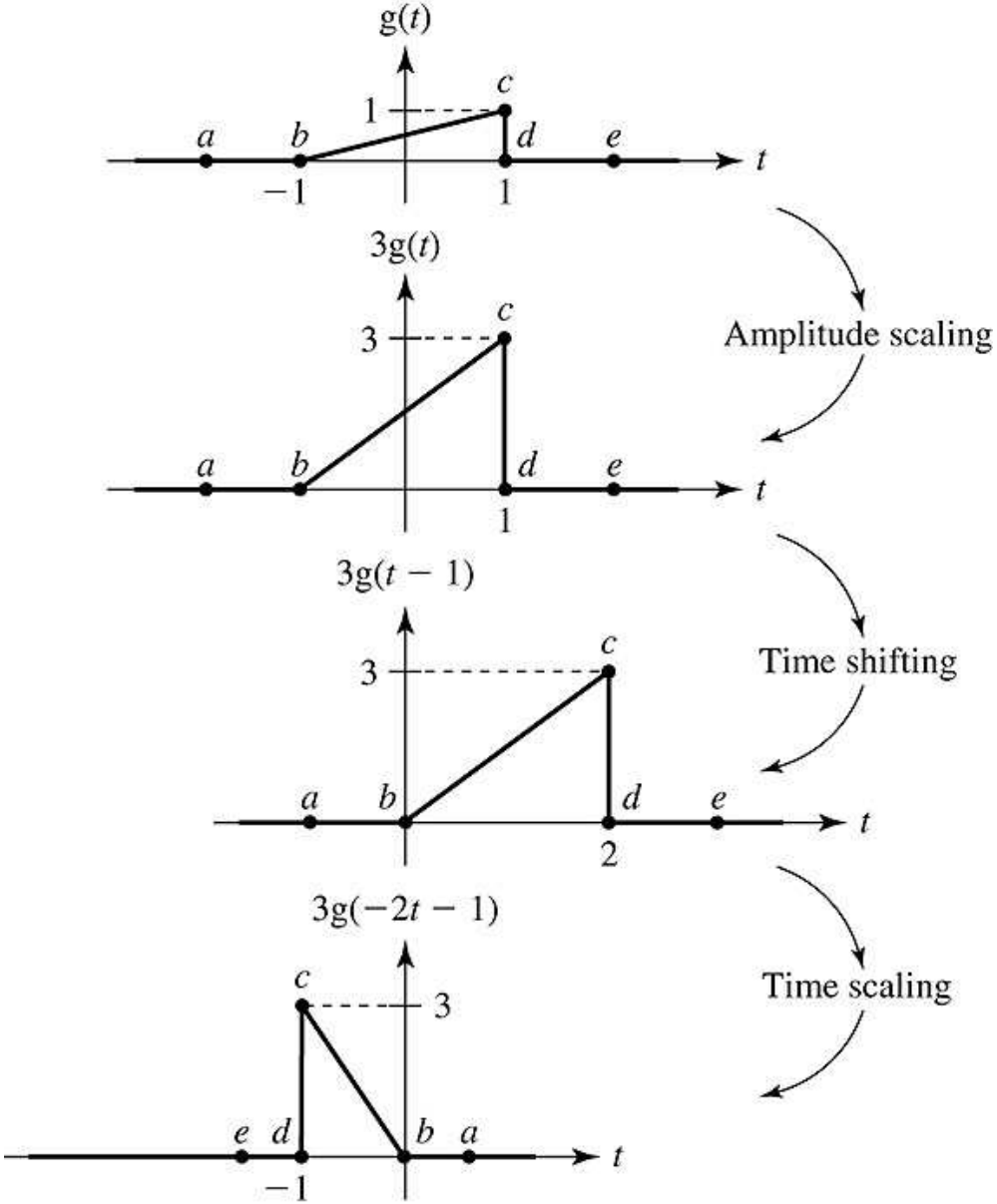
- In this case, time shifting must be performed before the time scaling:
  - Amplitude scaling: $g(t) \rightarrow Ag(t)$.
  - Time shifting $t \rightarrow t - t_0$: $Ag\left(t\right) \rightarrow Ag\left(t - t_0\right)$.
  - Time scaling $t \rightarrow bt$: $Ag(t - t_0) \rightarrow Ag\left(bt - t_0\right)$.
- Note that:

$$Ag\left(bt - t_0\right) = Ag\left(b(t - \frac{t_0}{b})\right)$$

  - Amplitude scaling: $g(t) \rightarrow Ag(t)$.
  - Time scaling $t \rightarrow bt$: $Ag(t) \rightarrow Ag\left(bt\right)$.
  - Time shifting $t \rightarrow t - t_0/b$: $Ag\left(bt\right) \rightarrow Ag\left(b(t - t_0/b)\right)$.

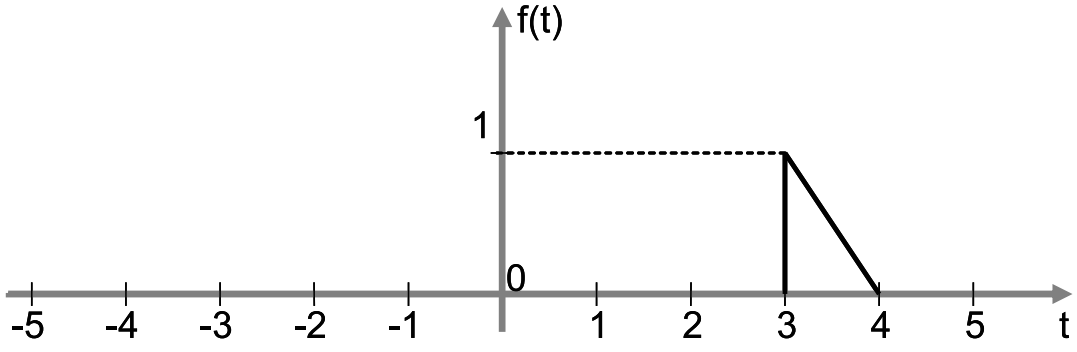
Multiple Transformations (4)

- The next figure corresponds to $g(t) \rightarrow 3g(-2t-1)$:



Example (1)

Consider the function $f(t)$ shown in the figure below. Plot $f(-4t+1)$ and $f(-2(t+3))$.

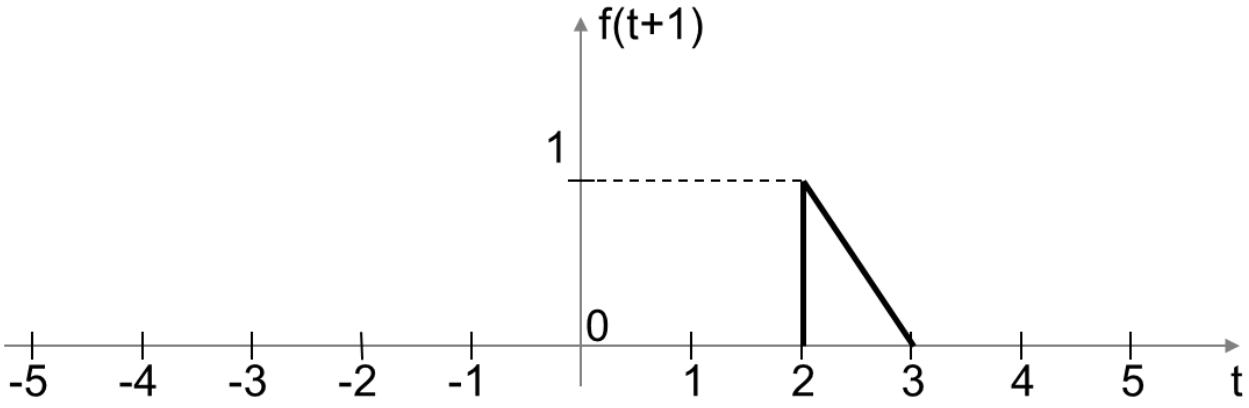


Example (2)

- The function $f(-4t+1)$ can be obtained by performing the following transformations:

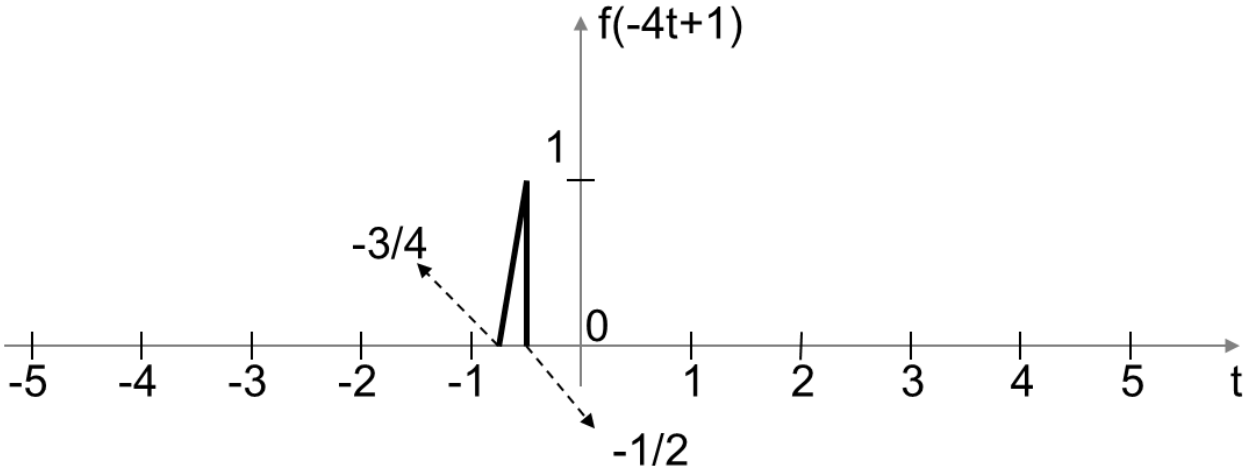
$$t \to t+1: \quad f(t) \to f(t+1)$$
$$t \to -4t: \quad f(t+1) \to f(-4t+1)$$

- $t \to t+1: \quad f(t) \to f(t+1)$:



Example (3)

- $t \to -4t: \quad f(t+1) \to f(-4t+1)$:



- Note that the duration of $f(t)$ is 1 while the duration of $f(-4t+1)$ is $1/4$ (the time-scaling factor is 4).

Example (4)

- Note that $f(t)$ can be written as:

$$f(t) = -t + 4 \quad ; \quad 3 \leq t \leq 4$$

- Consequently:

$$f(-4t+1) = -(-4t+1) + 4 = 4t + 3$$

- The limits of $f(-4t+1)$ can be determined from:

$$\begin{aligned} & 3 \leq -4t + 1 \leq 4 \\ \Rightarrow \quad & -4t + 1 \geq 3 \quad \Rightarrow \quad 4t \leq -2 \quad \Rightarrow \quad t \leq -1/2 \\ \Rightarrow \quad & -4t + 1 \leq 4 \quad \Rightarrow \quad 4t \geq -3 \quad \Rightarrow \quad t \geq -3/4 \end{aligned}$$

- Finally, $f(-4t+1)$ is given by:

$$f(-4t+1) = 4t + 3 \quad ; \quad -3/4 \leq t \leq -1/2$$

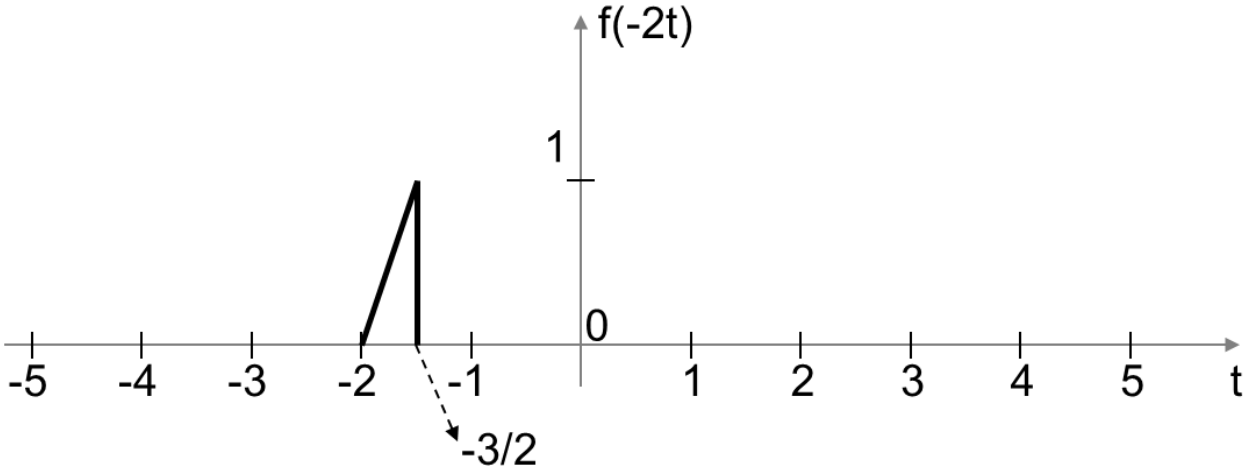
- This function corresponds to the plot that has been determined before.

Example (5)

- The function $f(-2(t+3))$ can be obtained by performing the following transformations:

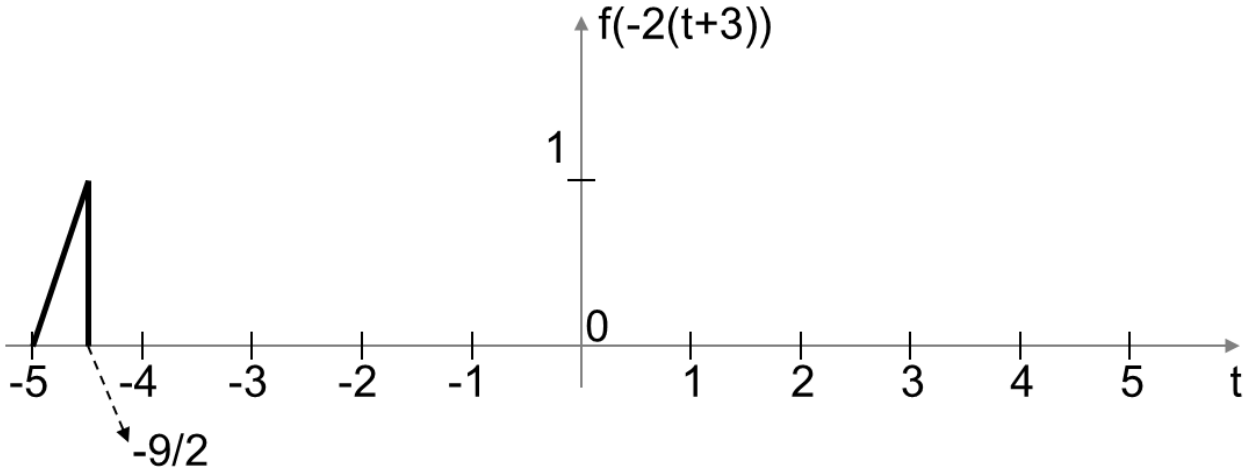
$$t \to -2t: \quad f(t) \to f(-2t)$$
$$t \to t+3: \quad f(-2t) \to f(-2(t+3))$$

- $t \to -2t: \quad f(t) \to f(-2t)$:



Example (6)

- $t \to t+3: \quad f(-2t) \to f(-2(t+3))$:



- Note that the duration of $f(t)$ is 1 while the duration of $f(-2(t+3))$ is $1/2$ (the time-scaling factor is 2).

Example (7)

- Note that $f(t)$ can be written as:

$$f(t) = -t + 4 \quad ; \quad 3 \leq t \leq 4$$

- Consequently:

$$f(-2(t+3)) = -(-2(t+3)) + 4 = 2t + 10$$

- The limits of $f(-2(t+3))$ can be determined from:

$$\begin{aligned} & 3 \leq -2(t+3) \leq 4 \\ \Rightarrow \quad & -2(t+3) \geq 3 \quad \Rightarrow \quad t+3 \leq -3/2 \quad \Rightarrow \quad t \leq -9/2 \\ \Rightarrow \quad & -2(t+3) \leq 4 \quad \Rightarrow \quad t+3 \geq -2 \quad \Rightarrow \quad t \geq -5 \end{aligned}$$

- Finally, $f(-4t+1)$ is given by:

$$f(-4t+1) = 2t + 10 \quad ; \quad -5 \leq t \leq -9/2$$

- This function corresponds to the plot that has been determined before.

Continuous-Time Even and Odd Functions (1)

- A CT even function verifies the relation:

$$g(t) = g(-t)$$

  Even functions are symmetrical about the y'y axis.

- A CT odd function verifies the relation:

$$g(t) = -g(-t)$$

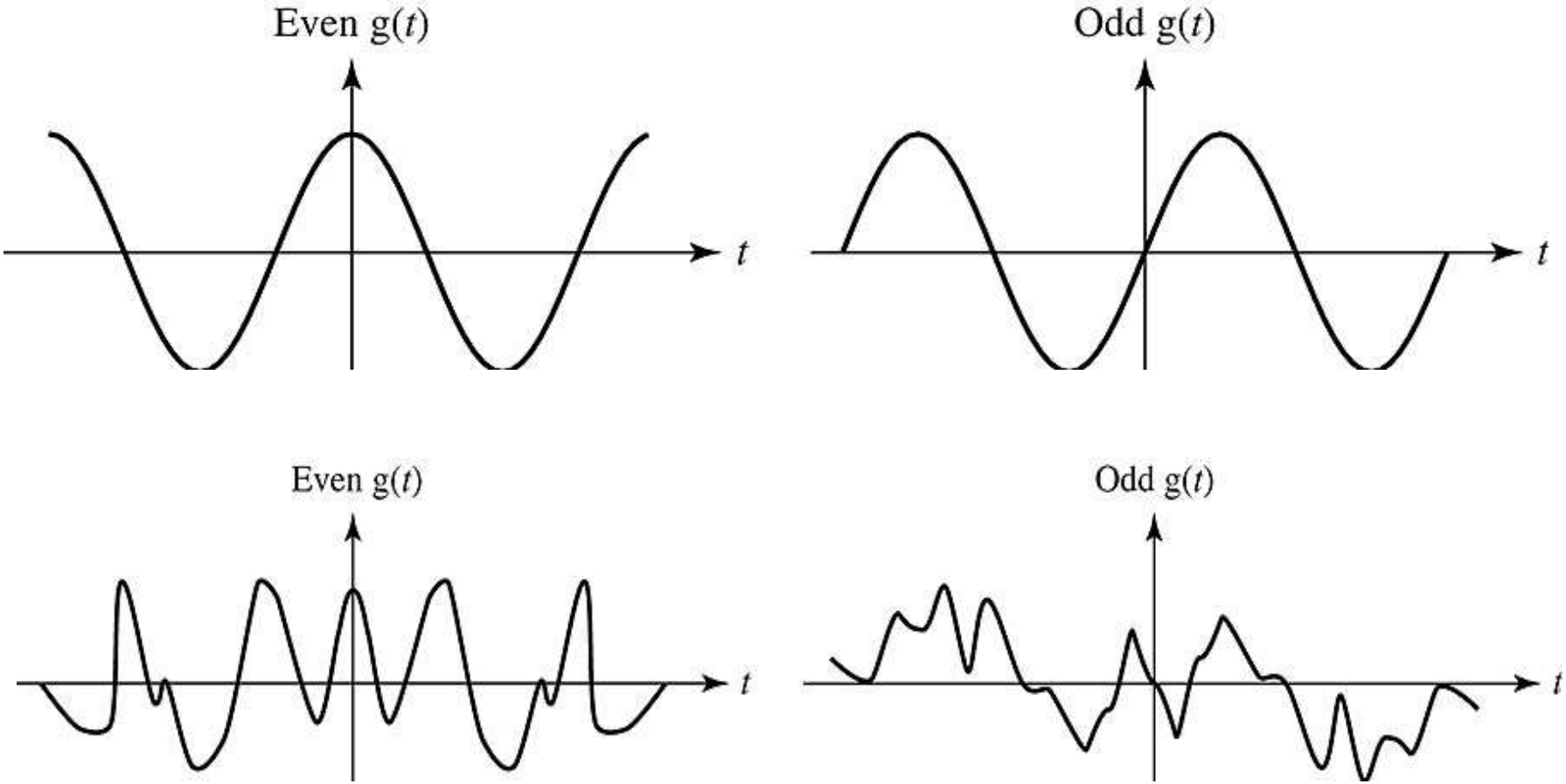
  Odd functions are symmetrical about the origin O.

- For example:
  - $g(t) = \cos(t)$ is even since $\cos(-t) = \cos(t)$.
  - $g(t) = \sin(t)$ is odd since $\sin(-t) = -\sin(t)$.
  - $g(t) = \mathrm{sinc}(t)$ is even since:

$$\mathrm{sinc}(-t) \triangleq \frac{\sin(-\pi t)}{-\pi t} = \frac{-\sin(\pi t)}{-\pi t} = \frac{\sin(\pi t)}{\pi t} = \mathrm{sinc}(t)$$

Continuous-Time Even and Odd Functions (2)

- The following figure shows examples of even and odd functions.



Continuous-Time Even and Odd Functions (3)

- Even functions verify the relation:

$$\int_{-a}^{a} g(t)\mathrm{d}t = 2\int_{0}^{a} g(t)\mathrm{d}t$$

- Proof:

$$\begin{aligned}
\int_{-a}^{a} g(t)\mathrm{d}t &= \int_{-a}^{0} g(t)\mathrm{d}t + \int_{0}^{a} g(t)\mathrm{d}t \\
&= -\int_{a}^{0} g(-t)\mathrm{d}t + \int_{0}^{a} g(t)\mathrm{d}t \\
&= \int_{0}^{a} g(-t)\mathrm{d}t + \int_{0}^{a} g(t)\mathrm{d}t \\
&= \int_{0}^{a} g(t)\mathrm{d}t + \int_{0}^{a} g(t)\mathrm{d}t = 2\int_{0}^{a} g(t)\mathrm{d}t
\end{aligned}$$

  - The second equality follows from the change of variable $t \to -t$ in the first integral.
  - The fourth equality follows since: $g(-t) = g(t)$.

Continuous-Time Even and Odd Functions (4)

- Odd functions verify the relation:

$$\int_{-a}^{a} g(t)\mathrm{d}t = 0$$

- Proof:

$$\begin{aligned}
\int_{-a}^{a} g(t)\mathrm{d}t &= \int_{-a}^{0} g(t)\mathrm{d}t + \int_{0}^{a} g(t)\mathrm{d}t \\
&= -\int_{a}^{0} g(-t)\mathrm{d}t + \int_{0}^{a} g(t)\mathrm{d}t \\
&= \int_{0}^{a} g(-t)\mathrm{d}t + \int_{0}^{a} g(t)\mathrm{d}t \\
&= -\int_{0}^{a} g(t)\mathrm{d}t + \int_{0}^{a} g(t)\mathrm{d}t = 0
\end{aligned}$$

  - The second equality follows from the change of variable $t \to -t$ in the first integral.
  - The fourth equality follows since: $g(-t) = -g(t)$.